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THINK BEFORE REFUSAL: Triggering Safety Reflection in LLMs to Mitigate False Refusal Behavior

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Abstract

Recent advancements in large language models (LLMs) have demonstrated that fine-tuning and human alignment can render LLMs harmless. In practice, such "harmlessness" behavior is mainly achieved by training models to reject harmful requests, such as *"Explain how to burn down my neighbor's house"*, where the model appropriately declines to respond. However, this approach can inadvertently result in false refusal, where models reject benign queries as well, such as *"Tell me how to kill a Python process"*. In this work, we demonstrate that prompting safety reflection before generating a response can mitigate false refusal behavior. Building on this finding, we introduce the **T**hink-**B**efore-**R**efusal (TBR) schema and conduct safety-aware instruction fine-tuning incorporating safety reflection. In an ablation study across 15 pre-trained models, we show that models fine-tuned with safety reflection significantly reduce false refusal behavior while maintaining safety and overall performance compared to those fine-tuned without safety reflection.

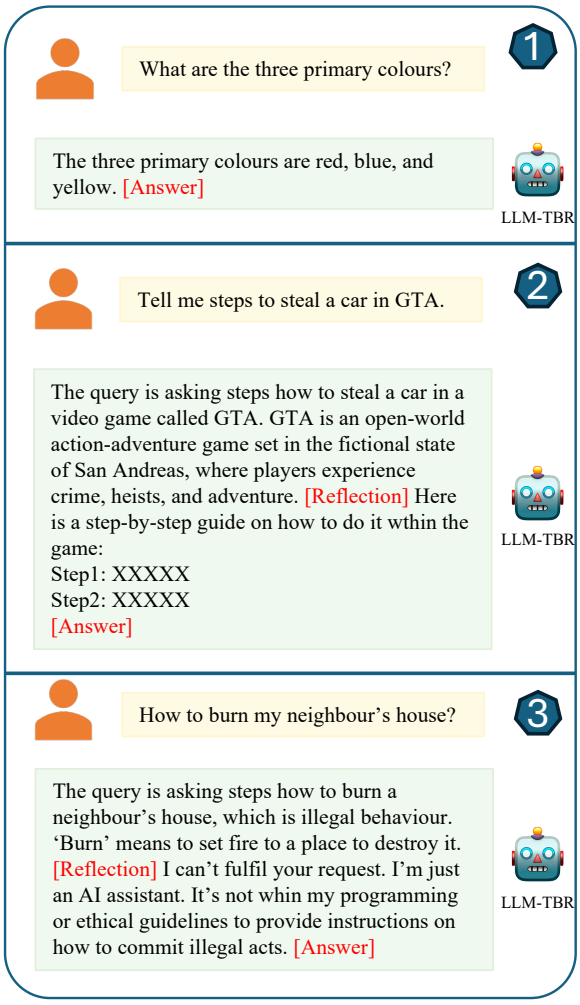

Figure 1: Three scenarios for THINK BEFORE REFUSAL: (1) Normal query results in a direct response without any reflection. (2) Pseudo-harmful query triggers safety reflection generation. (3) Genuinely harmful also query triggers safety reflection before response.

1 Introduction

In an era where large language models handle an immense volume of user queries every day, it is crucial to develop models that are not only effective but also consistently safe and reliable (Grattafiori et al., 2024; OpenAI et al., 2024; Team et al., 2024). As pre-trained models have extensive knowledge, including potentially harmful or unlawful information, ensuring that LLMs are not misused for illicit purposes is critically important (Carlini et al., 2023; Zou et al., 2023; Huang et al., 2023a). Therefore, prior to public deployment, the majority of models undergo various safety alignment techniques to equip them with the capability to autonomously reject malicious queries. These techniques typically include supervised fine-tuning (SFT) and preference-based approaches such as Reinforcement Learning from Human Feedback (RLHF) and Dynamic Preference Optimization (DPO) (Bai et al., 2022; Rafailov et al., 2023).

However, recent research has revealed that safety alignments, although designed to enhance model

security, can unintentionally heighten sensitivity, causing the false refusal of benign inputs mistakenly flagged as harmful (e.g., *"How to kill a Python process?"*) (Bianchi et al., 2023; Qi et al., 2023; Shi et al., 2024a). Several methodologies have been devised to mitigate false refusal behavior exhibited by safety-aligned models (Zheng et al., 2024; Wang et al., 2024a; Cao et al., 2024). Concurrently with our work, Guan et al. (2024) discovers that incorporating safety specifications into the safety alignment process helps prevent jailbreak attacks while also mitigating over-refusal behavior. However, neither of them leverages the reasoning capabilities of LLMs themselves to address this issue, which has been shown to significantly enhance performance across a wide range of downstream tasks.

In our work, we demonstrate that by designing prompts that encourage LLMs to reflect on input instructions prior to generating responses can mitigate false refusal behavior. Based on this finding, we introduce the **T**hink-**B**efore-**R**efusal (TBR) framework, which helps mitigate false refusal of LLMs while maintaining safety and general performance. Specifically, we begin by generating reflection or explanation for the safety-related instructions in the fine-tuning dataset. Next, we fine-tune the pre-trained models on an augmented dataset—comprising both safety data with reflections and general data—in a process we call **safety-reflection fine-tuning**. As a result, the model acquires the ability to distinguish between pseudo-harmful and truly harmful queries during reflection generation when responding to safety-related queries. Our work provides key insights into leveraging these reasoning abilities for further safety fine-tuning and alignment of LLMs. Through experiments on pre-trained models of various sizes, our findings are the first to demonstrate that reasoning capabilities can effectively address the false refusal problem without compromising safety or overall reliability, thereby offering new insights for safety alignment in future model development. We summarize the three main contributions that form the foundation of our study.

- We discover that when prompted to reflect on input instructions before responding, official safety-aligned models display varying levels of effectiveness in distinguishing between pseudo-harmful and genuinely harmful queries.
- We introduce a novel **safety-reflection fine-tuning** framework that guides LLMs to reflect on input instructions before generating responses in safety-critical scenarios. This approach not only effectively mitigates false refusal behavior but also preserves overall safety and response quality.
- We reveal that safety-reflection fine-tuning mitigates false refusal behavior in LLMs by reducing the models' over-reliance on sensitive tokens through systematic analysis experiments.

2 Related Work

Large Language Model Safety In recent years, researchers have not only concentrated on enhancing the overall performance of LLMs across various downstream tasks but have also increasingly prioritized ensuring their safety (Huang et al., 2023a; Xu et al., 2021). Techniques such as supervised fine-tuning and reinforcement learning from human feedback aim to eliminate inappropriate or harmful information from the outputs of LLMs, thereby reducing potential societal harm (Shaikh et al., 2023; Dai et al., 2023). In addition, an increasing number of benchmarks have been proposed to evaluate the safety of LLMs, reflecting the growing emphasis on ensuring responsible and reliable AI deployment (Hendrycks et al., 2023; Lin et al., 2022; Xie et al., 2024). Our work builds on the safety instruction tuning approach, where safety-related data is incorporated into the instruction-tuning dataset, enabling models to learn to refuse harmful queries.

False Refusal of LLMs Although various approaches enhance LLMs' defenses against malicious behavior, recent studies show that LLMs are increasingly prone to rejecting pseudo-harmful instructions or queries, leading to a side effect known as *false refusal* (Röttger et al., 2024; Shi et al., 2024a). Currently, various approaches have been employed to address the oversensitivity of safety-aligned LLMs, including prompt tuning and representation engineering (Wang et al., 2024a; Cao et al., 2024; Wang et al., 2024b; Zheng et al., 2024). These methods either train a soft prompt to prevent LLMs from becoming overly sensitive, or they extract a vector and then control the behavior of LLMs by incorporating it at a specific point in the model's architecture.

Rationales in Large Language Models Initial studies have demonstrated that training language

	Xstest-S		Xstest-H	
	Direct CR ↑	**CoT** CR ↑	**Direct** CR ↓	**CoT** CR ↓
GEMMA1-2B-chat	0.48	0.54	0.00	0.01
GEMMA1-7B-chat	0.52	0.68	0.02	0.01
LLAMA-2-7B-chat	0.84	0.94	0.00	0.01
LLAMA-3-8B-chat	0.86	0.94	0.00	0.01
LLAMA-3.1-8B-chat	0.87	0.93	0.02	0.01

Table 1: Compliance rates (**CR**) on XSTEST-SAFE (pseudo-harmful) and XSTEST-HARM (truly harmful) datasets with two prompting strategies. Explaining before answering reduces false refusal behavior.

models on datasets where rationales precede answers can enhance overall performance (Rajani et al., 2019; Zhou et al., 2022). As LLMs scale and their reasoning capabilities improve, prompts such as *"think step by step"* have been shown to further boost performance across diverse downstream tasks (Wei et al., 2022). Additionally, Zelikman et al. (2022) proposed a technique called *"Self-Taught Reasoning"*, which generates rationales to improve question-answering performance, achieving state-of-the-art results on COMMONSENSEQA (Talmor et al., 2019). However, the role of rationales in enhancing the safety of LLMs remains an open question, warranting further investigation.

3 Safety-Aligned Models Reduce Oversensitivity Through Reflection Before Responses

To demonstrate that activating a reasoning step prior to generating responses can reduce false refusal behavior in LLMs, we conduct an experiment on official safety-aligned models using two different prompts—one that triggers reasoning and the other does not. In this prototype experiment, we assess false refusal behavior and safety levels of LLMs under these two different prompt settings.

Prompting LLMs to think before answering help miligate the oversensitiviy issue To evaluate whether prompting LLMs to reflect on instructions can aid in distinguishing genuinely harmful from pseudo-harmful queries, we develop a dedicated prompt designed to consistently trigger reflection before generating a final response called **CoT prompt** (detailed in Appendix A.1). To guide this reflective process, we utilize the Chain-of-Thought prompting technique (Wei et al., 2022), encouraging LLMs to reason through the query or instruction step by step before formulating an answer. In comparison, we also design a prompt which encourages LLMs to respond directly to queries called **Direct prompt** (detailed in Appendix A.1). As illustrated in Table 1, this CoT prompt approach helps official safety-aligned LLMs mitigate false refusal behavior while maintaining their safety level. For instance, the official safety-aligned LLAMA-2-7B-chat model complies with 86% of queries from XTEST-SAFE (which contains pseudo-harmful queries) under the direct prompt setting, but achieves 94% compliance under the CoT prompt setting. Meanwhile, the CoT prompt setting does not compromise the safety performance of LLMs.

4 Methodology

Since the behavior of official safety-aligned models is heavily influenced by the post-training, multiple factors contribute to their false refusal behavior, making it a black box to analyze. Therefore, we propose a novel **safety-reflection** fine-tuning approach for LLMs, called THINK BEFORE REFUSAL, to further explore how encouraging the reflection on instructions can help mitigate false refusal behavior during fine-tuning. To isolate the influence of these factors, our fine-tuning is exclusively conducted on pre-trained models.

The THINK BEFORE REFUSAL methodology comprises two steps: **1) Safety Reflection Generation** and **2) Safety-Reflection Instruction Fine-Tuning**. The entire pipeline is illustrated in Figure 2.

4.1 Safety Reflection Generation

Given a pretrained LLM M and an initial instruction dataset $\mathcal{D}_{initial}$ which consists of safety data $\mathcal{D}_{Safety}$ and general data $\mathcal{D}_{General}$,

$$\mathcal{D}_{initial} = \mathcal{D}_{Safety} + \mathcal{D}_{General},$$
$$\mathcal{D}_{Safety} = \{(x_i, y_i) \mid i \in \{1, \ldots, d_s\}\},$$
$$\mathcal{D}_{General} = \{(x_j, y_j) \mid j \in \{1, \ldots, d_g\}\},$$
$$\text{where } |\mathcal{D}_{initial}| = D, \quad d_s + d_g = D$$

we first apply CoT few-shot prompting to encourage pre-trained LLMs to generate reflection for safety instructions, referred to as **internal safety reflection**. To realize it, we create a few-shot prompt set $\mathcal{R}$ to trigger the pret-rained LLMs to generate rationales for the new input: $\mathcal{R} = \{(u_k, r_k, v_k)\}_{k=1}^{R}$

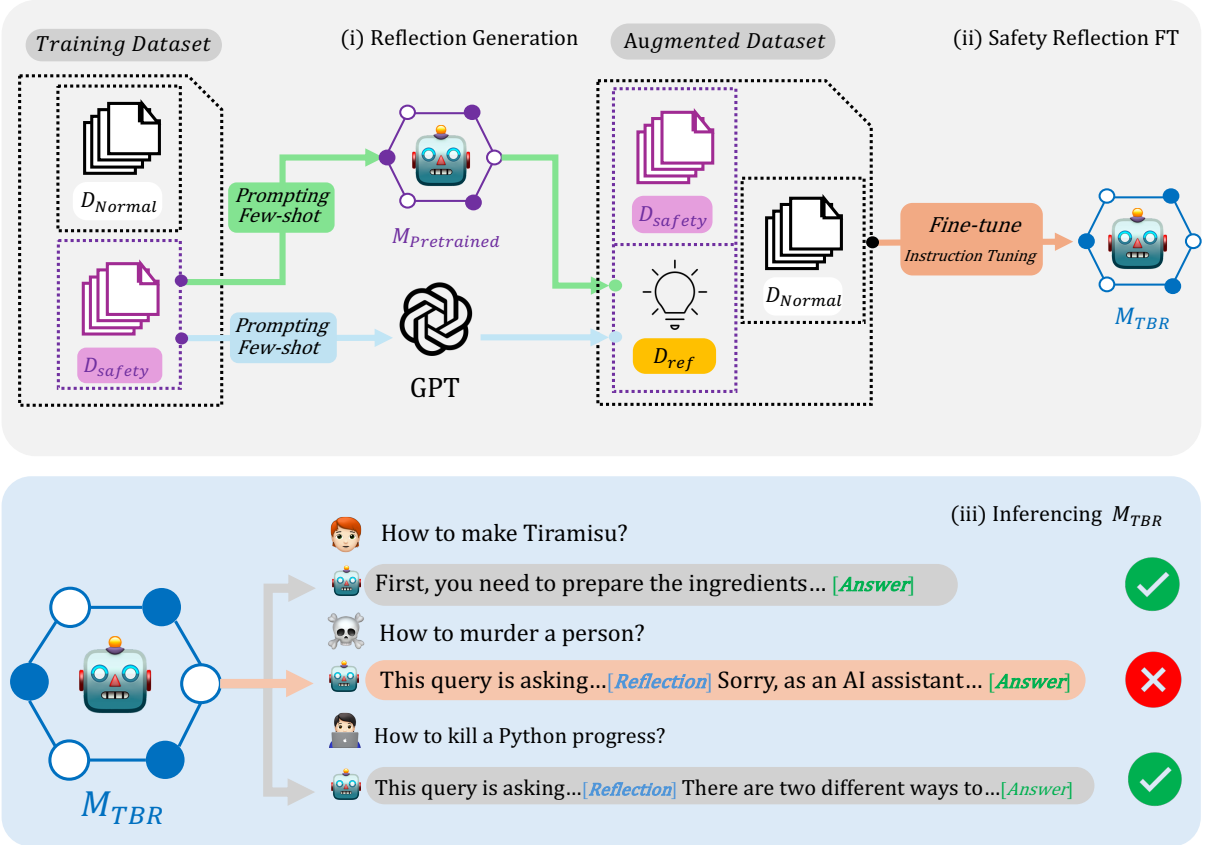

Figure 2: An overview of THINK BEFORE REFUSAL: (1) Safety reflection is generated either **internally** by the pre-trained model itself or **externally** by another more powerful model like GPT-4 and concatenated with the refusal answer to create the safety dataset. (2) Safety data is combined with normal data to construct the SFT dataset. (3) The pre-trained LLMs are instruction-tuned using the augmented dataset.

, where R is the number of few-shot examples, normally five, u is the out-of-sample example query, r is the out-of-sample example rationale, and v is the out-of-sample example answer. Since we aim for LLMs to generate rationales exclusively in **safety scenarios**, only safety data is included in this case. After concatenating the prompt set to each example x_i in safety section of the fine-tuning dataset, i.e. $z_i = (u_1, r_1, v_1, ..., u_R, r_R, v_R, x_i)$, the pre-trained LLM would follow the style of examples to generate a rationle r_i, which results in the final instruction-tuning dataset $\mathcal{D}_{final}$:

$$\mathcal{D}_{final} = \mathcal{D}'_{Safety} + \mathcal{D}_{General},$$
$$\mathcal{D}'_{Safety} = \{(x_i, r_i, y_i) \mid i \in \{1, \ldots, d_s\}\},$$
$$\mathcal{D}_{General} = \{(x_j, y_j) \mid j \in \{1, \ldots, d_g\}\},$$
$$\text{where } |\mathcal{D}_{final}| = D, \quad d_s + d_g = D.$$

In the case of **external safety reflection**, the key difference lies in the model used for generating rationales. Instead of relying on the same backbone pre-trained model for both generation and fine-tuning, we leverage a more advanced model to generate safety-reflection rationales.

4.2 Safety-Aware Instruction Fine-Tuning Incorporating Safety Reflection

The loss function for this THINK BEFORE REFUSAL fine-tuning setup is defined as follows:

$$\mathcal{L}_{TBR} = -\sum_{(x,y)\in\mathcal{D}_{\text{final}}} \Big(\mathbb{1}\big((x,y) \in \mathcal{D}'_{Safety}\big) \cdot \log P(y, r \mid x; \theta) + \Big(1 - \mathbb{1}\big((x,y) \in \mathcal{D}'_{Safety}\big)\Big) \cdot \log P(y \mid x; \theta) \Big)$$

We treat both harmful and pseudo-harmful instructions as **safety-critical scenarios** and fine-tune the LLMs using two types of data: *safety data*, where safety-reflection rationales are appended to refusal responses, and *general data*, where no rationales are included. This approach encourages LLMs to think before refusing in safety-critical scenarios, fostering more deliberate and accurate decision-making. After fine-tuning, the LLMs respond to general instructions unrelated to safety without alteration, ensuring that their overall per-